

Churn Prediction for FoodCorp

Technical Report — BUSI4373 Machine Learning and Predictive Analytics

Student ID: 20811152

Module: BUSI4373 — MSc Business Analytics

Date: May 14, 2026

I confirm that I used Large Language Model (LLM) assistance during this coursework. The use was strictly within the scope permitted by the coursework AI policy, as detailed below.

What AI was used for

- Debugging Python and SQL code errors (e.g. resolving Spark DataFrame schema mismatches, fixing matplotlib formatting issues, diagnosing Spark Decimal-to-float conversion errors).
- Looking up scikit-learn function signatures (e.g. permutation_importance parameters, ParameterGrid usage, HistGradientBoostingClassifier hyperparameter names).
- Improving figure and table clarity, including suggestions on axis labelling, legend placement, choice of normalisation for cross-feature comparison plots, and consistent colour-coding between churner and non-churner categories.
- Polishing the academic English of the final report draft and standardising terminology (e.g. consistent use of “reference date”, “observation window”, “output window”, “temporal holdout”).
- Helping structure the report sections to match the coursework brief and rubric weightings, without generating the report content itself.

What AI was NOT used for

- Choosing the churn definition or value of β — selected by me based on the ConsultingCorp report and my own empirical inter-visit gap analysis.
- Designing the temporal evaluation strategy or the windowing scheme — the choice of fixed train/validation/test reference dates, tumbling window width, observation-window length and label-window construction are my own analytical decisions, informed by the module lectures.
- Selecting the candidate models or final model — I designed the candidate set, implemented the manual grid search on the validation reference date, evaluated overfitting gaps and chose the tuned random forest based on its validation ROC-AUC and small overfit gap.
- Interpreting results — all interpretation of permutation importance, churner vs non-churner differences, pen portraits and operational threshold recommendations is my own.
- Generating any numerical results, performance metrics, feature values or figures — every figure was produced by code I wrote in the Databricks notebook and every metric in the report comes directly from that notebook's output cells.
- Writing the report end-to-end — AI was used as a writing aid only, not as an author. The structure, analytical reasoning, limitations discussion and conclusions are my own.

Tools used

Anthropic Claude (large-language-model assistant) was used through its web interface, primarily for the purposes listed above. No agentic system, autonomous coding agent or automated submission pipeline was used. No prompts were used to generate analytical outputs that appear in the report or notebook.

Statement of authorship

All analytical decisions, all results, all interpretation, and all final report content are my own. Where AI suggestions were incorporated into the wording of the report I reviewed, edited and verified them against my actual notebook outputs. I accept full responsibility for the content of this submission.

1. Executive Summary

This report develops a binary churn-prediction system for FoodCorp using two years of loyalty-card transactional data (8,526 customers, 200,780 receipts, 1,942,995 line items, four stores, 2020-07-22 to 2022-03-22). The business objective is to identify, at any chosen reference date R , which currently active customers are most likely to stop shopping with FoodCorp during the following 49 days so that targeted retention activity can be planned.

Following the ConsultingCorp report, churn is defined globally as $(R - \text{date_of_last_purchase}) > \beta$, with $\beta = 49$ days. This β corresponds to the upper-quartile customer median inter-visit gap; at this value 74.85% of customers are not, on average, considered to have churned, and ConsultingCorp's perfect-classifier analysis expects roughly 8.81% of active customers to be flagged on an average prediction day. A customer is treated as active at R if their most recent visit occurred within β days of R . At the validation reference date ($R = \text{day } 400$, $\approx$ August 2021) the forward-looking churn rate among active customers is 29.2%.

Features were engineered predominantly in SQL on Databricks, producing 30 candidate predictors across six conceptual groups: recency, four 14-day tumbling spend windows, equivalent visit and quantity windows, derived spend trends and ratios, 56-day breadth measures (distinct products, categories, sub-categories, departments, stores), and lifetime customer features (total visits, total spend, mean inter-visit gap, tenure). All feature windows are computed strictly using transactions on or before R , with labels constructed from the subsequent β -day window. A direct SQL leakage check confirmed zero discrepancies between recorded last-purchase day and the corresponding recency feature.

Evaluation uses an explicit temporal holdout: a fixed train reference date ($R_{\text{TRAIN}} = 300$, $\approx$ May 2021), a fixed validation reference date used for meta-parameter tuning ($R_{\text{VAL}} = 400$), and a fixed out-of-time test reference date ($R_{\text{TEST}} = 500$, $\approx$ December 2021). Random cross-validation and unmodified GridSearchCV / TimeSeriesSplit were deliberately avoided because they cannot enforce a strict past-to-future ordering in this single-cohort, panel-style data.

Four candidate classifiers were compared (logistic regression, decision tree, random forest, histogram-based gradient boosting), each tuned by manual grid search using the validation reference date only. The tuned random forest ($\text{max_depth} = 5$, $\text{min_samples_leaf} = 20$, no class weighting) was selected as the final model because it achieved the highest validation ROC-AUC (0.891) with the smallest train-validation gap (0.005), indicating well-controlled overfitting. Retrained on combined train+validation data and evaluated only once on the test reference date, the model achieves ROC-AUC = 0.872, PR-AUC = 0.702, accuracy = 0.808, precision = 0.751, recall = 0.449, $F1 = 0.562$ at the default 0.50 threshold. At a lowered 0.30 threshold, precision falls to 0.628 but recall rises to 0.708, flagging 30.9% of active customers — a trade-off appropriate for low-cost interventions such as email or app-push offers.

Permutation importance on the validation set identifies four lifetime-engagement features ($\text{total_visits_lifetime}$, $\text{total_spend_lifetime}$, avg_gap_lifetime , $\text{customer_tenure_days}$) as the dominant signals, followed by recent-trend features (spend_ratio_w1w2 , $\text{days_since_last_purchase}$). Pen portraits show a typical churning customer with 2 lifetime visits, £40.45 lifetime spend and a 23-day median inter-visit gap, against a non-churner with 15 lifetime visits, £261.85 lifetime spend and a 12-day gap. Churn risk is therefore concentrated in low-engagement and newer customers; building early engagement appears more leverage-rich than reacting to short-run dips.

The model is materially better than the strongest baseline ($\text{recency} > 7$ days: ROC-AUC = 0.728) and is interpretable enough for FoodCorp's in-house team to monitor monthly. Principal limitations are the single test reference date (repeated temporal holdout would strengthen stability evidence), the absence of an explicit promotion-response feature, the very low store count, and the high missingness of age (87%) which forced the demographic to be dropped.

2. Current Levels of Churn

ConsultingCorp's report formally defines churn as $(date_today - date_of_last_purchase) > \beta$, deliberately treating the calendar-day difference rather than the time of day or individual visit frequency. β is the single remaining free parameter, and ConsultingCorp's two diagnostic curves — the cumulative distribution of customer median inter-visit gaps and the percentage of customers who churn under a perfect predictor (PCCC) — are intended to bound a reasonable choice rather than dictate one.

A fixed value of $\beta = 49$ days was selected. Three reasons justify this. First, as shown in Figure 1, the distribution of customer median inter-visit gaps has a median of 18 days and an upper quartile of 50 days. Choosing $\beta = 49$ therefore places the churn threshold just below the P75 shopping cadence, meaning a customer is only labelled as churned after a gap that is genuinely long for most FoodCorp customers. Second, ConsultingCorp's Figure 3 shows that at $\beta = 49$ the perfect-classifier PCCC is 8.81%, which equates to ~25 customers per prediction day on average — an operationally sensible target volume for a four-store retailer running periodic retention campaigns. Third, a β sensitivity check across $\{18, 26, 30, 35, 40, 49\}$ at $R = 400$ shows that, on the active cohort, $\beta = 49$ yields the largest usable population (3,641 active customers) and the most balanced forward-looking churn rate (29.2%), avoiding the over-aggressive labelling that $\beta = 18$ would imply (37.3% churn).

Table 1 — β sensitivity at $R = 400$

β (days)	Active customers	Churners	Non-churners	Churn rate	Retention rate
18	2,437	910	1,527	37.3%	62.7%
26	2,873	974	1,899	33.9%	66.1%
30	3,016	984	2,032	32.6%	67.4%
35	3,179	989	2,190	31.1%	68.9%
40	3,346	1,013	2,333	30.3%	69.7%
49	3,641	1,062	2,579	29.2%	70.8%

Under this definition, an active customer at any reference date R is one whose most recent transaction occurred no more than 49 days before R . Already-churned customers — those whose last visit predates R by more than 49 days — are excluded from the modelling cohort because they have already entered FoodCorp's stop-targeting list, in line with the operational rule set out in the ConsultingCorp report. At $R = 400$ this leaves 3,641 active customers, of which 1,062 fail to return within the following 49 days (29.17% forward-looking churn). The rate is also reasonably stable across reference dates — 29.5% at $R = 300$, 29.2% at $R = 400$ and 27.4% at $R = 500$ — which is later summarised visually in Figure 3.

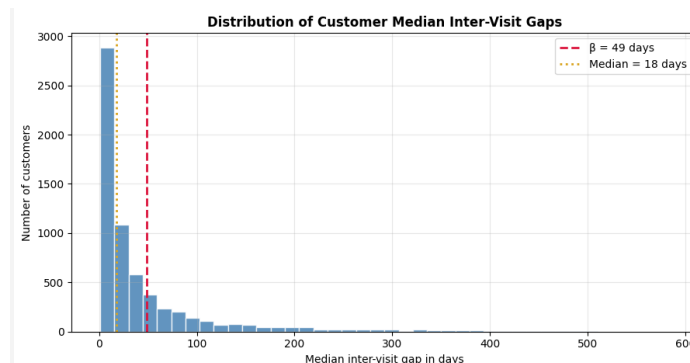


Figure 1. Distribution of customer median inter-visit gaps with $\beta = 49$ marked. P50 = 18 days, P75 = 50 day

3. Technical Report on the Prediction System

This section describes the prediction system as an end-to-end technical pipeline: feature construction, model selection, evaluation, and interpretation safeguards. It focuses on design choices needed to replicate the system, rather than routine implementation steps.

3.1 Features

Feature engineering was carried out predominantly in SQL through a single parameterised Databricks function `build_feature_sql(R,  $\beta$ , w_size, n_windows)` that emits a CTE pipeline for any chosen reference date. This guarantees that the train, validation and test feature matrices are produced by identical logic, removing the risk of implementation drift between splits. A flat view `ml34.v_transactions` is the single source joining the five raw tables, with a clean value column that nulls the 9,999,999 sentinel observed in `receipt_lines`.

Windowing strategy. The full feature and label setup is summarised in Figure 2. Each reference date R defines a 56-day observation window, split into four non-overlapping 14-day tumbling windows. Window w_1 is the most recent period, $(R-14, R]$, followed by w_2 , w_3 and w_4 moving backwards in time. The label window is then placed strictly after R , covering $(R, R+\beta]$. A customer is labelled as churned if they do not purchase again during this 49-day forward-looking period. **Strict inequalities at the right edge of feature windows and at the left edge of the label window** prevent the same calendar day from appearing in both, eliminating leakage across the reference date.

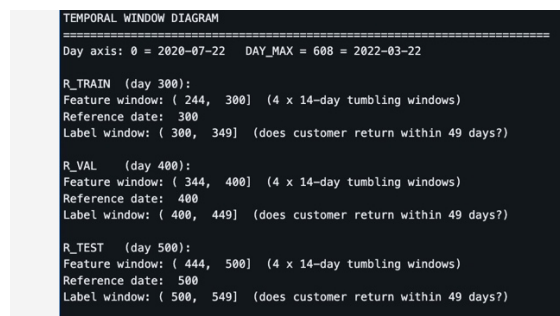


Figure 2. Windowing strategy. Per reference date R , four 14-day tumbling input windows (w_4 – w_1) produce features; the 49-day output window after R produces the label. Train, validation and test splits use disjoint R values.

Feature groups tried. Six conceptual groups, totalling 30 retained predictors, were constructed:

- Recency: `days_since_last_purchase`.
- Spend windows: `spend_w1`–`w4` (sum of cleaned line value).
- Visit windows: `visits_w1`–`w4` (distinct purchase days per window, not receipt counts).
- Quantity windows: `qty_w1`–`w4` (sum of line quantities).
- Basket / intensity: `avg_basket_w1`, `avg_basket_w2`, `avg_items_per_visit_w1`.
- Trends/ratios: `spend_trend_w1w2`, `spend_trend_w1w3`, `visit_trend_w1w2`, `visit_trend_w1w3`, `spend_ratio_w1w2` (captures spend decline between adjacent windows).
- Breadth over the 56-day observation window: `n_products_56d`, `n_categories_56d`, `n_subcategories_56d`, `n_departments_56d`, `n_stores_56d`.
- Lifetime tenure features: `total_visits_lifetime`, `total_spend_lifetime`, `avg_gap_lifetime`, `customer_tenure_days`.
- Demographic: `age_at_R`, computed as $\text{DATEDIFF}(R, \text{dob}) / 365.25$.

Features kept and discarded. `age_at_R` was discarded because 86–88% of rows were null across all three splits, making any imputed signal more noise than information. All other 30 features were retained for modelling. Permutation importance on the validation set produced no features with importance below -0.001 , so no further drop was applied. A reduced-feature variant (features only with importance ≥ -0.001) was tested; it lost 0.0004 ROC-AUC, well within the 0.002 tolerance defined a priori, so the full feature set was retained for parsimony of the tracked grouped scaling.

Preprocessing and scaling. A bespoke temporal-group scaler was implemented rather than per-column `StandardScaler`. All four spend windows share a single (mean, std) computed from the pooled w_1 – w_4 training values; the same logic is applied to visit, quantity, basket, distinct-product and distinct-category window groups. This preserves the magnitude relationships between adjacent tumbling windows (e.g. `spend_w1 < spend_w2` retains meaning after scaling), which would be lost under independent column scaling. All scaling statistics are computed on the training split only and applied unchanged to validation and test.

Leakage prevention. A SQL cross-check compared, for every customer active at $R = 300$, the true $\text{MAX}(\text{day}) \leq R$ from `v_transactions` against the `days_since_last_purchase` feature: 0 discrepancies, maximum discrepancy 0. Feature CTEs use `t.day ≤ R` as the WHERE filter; label CTEs use `day > R AND day ≤ R + β`. The three splits use disjoint reference dates ($R = 300, 400, 500$) and the test reference is 100 days beyond validation, so no test customer's input window overlaps with validation labels.

3.2 Prediction Approach

The pipeline follows the sequence: SQL feature/label tables per R → Spark → pandas load → age imputation (median = 52.1 years, although the column is ultimately dropped) → temporal-group scaling fitted on train only → candidate classifier fit on train → validation-set evaluation → manual hyper-parameter search → final retraining on train+validation → one-shot evaluation on the held-out test reference date. Four candidate algorithms were compared: logistic regression (linear baseline), decision tree (fully overfit by design, to characterise the upper bound and motivate regularisation), random forest (bagged ensemble with regularisable depth) and `HistGradientBoostingClassifier` (sequential ensemble with built-in early stopping support).

Two simple business baselines anchor the comparison. Baseline 1 predicts every active customer as non-churn; with a 29% positive rate this gives 70.8% validation accuracy but zero recall and is operationally useless. Baseline 2 sweeps a threshold on `days_since_last_purchase` and selects the validation-best value (recency > 7 days), giving $F1 = 0.557$ and $\text{ROC-AUC} = 0.728$.

All candidate classifiers were first fitted with library defaults to surface overfitting. The decision tree and random forest reached perfect training AUC (1.000), and HGBT reached 0.999, against validation AUCs of 0.685, 0.877 and 0.867 respectively. Logistic regression, by contrast, produced near-identical train and validation AUC (0.876 vs 0.882). This pattern motivated explicit regularisation for the three tree-based families during meta-parameter tuning.

Meta-parameter tuning. Standard scikit-learn GridSearchCV and TimeSeriesSplit were avoided because they cannot express the requirement that an entire customer's history before R becomes input and an entire customer's behaviour after R becomes label. Instead, a manual ParameterGrid loop fits every grid candidate on the train reference date and scores it on the validation reference date — a temporal holdout that respects the panel structure. Grids were:

- Logistic regression: $C \in \{0.01, 0.1, 1, 10\}$, $\text{class_weight} \in \{\text{None}, \text{balanced}\}$, $\text{penalty} = 12$.
- Random forest: $\text{max_depth} \in \{5, 10, 15, \text{None}\}$, $\text{min_samples_leaf} \in \{5, 10, 20\}$, $\text{class_weight} \in \{\text{None}, \text{balanced}\}$.
- HGBT: $\text{learning_rate} \in \{0.05, 0.1, 0.2\}$, $\text{max_depth} \in \{3, 5, \text{None}\}$, $\text{min_samples_leaf} \in \{10, 20, 40\}$.

The tuned random forest ($\text{max_depth} = 5$, $\text{min_samples_leaf} = 20$, $\text{class_weight} = \text{None}$) was selected as the final model. It produced the highest validation ROC-AUC (0.8914) and the smallest train-validation gap (0.0046), against HGBT's 0.0275 gap. The $\text{class_weight} = \text{'balanced'}$ configurations consistently raised F1 by lifting recall but reduced ROC-AUC by 0.0010–0.0020 and produced more variable predictions across thresholds, so the unweighted model was preferred and class imbalance was instead handled via the operating-threshold discussion in §3.3.

3.3 Evaluation

The temporal evaluation design is shown in Figure 3. Model selection is based only on the validation reference date, while the test reference date is held back and used once after the final model and meta-parameters have been fixed. This follows a single-period out-of-time evaluation strategy and reduces the risk of procedural overfitting. Random k-fold cross-validation would mix future-labelled customer behaviour into training, while TimeSeriesSplit on a flat customer table would not respect the customer-level panel structure.

Five metrics are reported because no single number captures the trade-offs of a 29% imbalanced classification problem. ROC-AUC measures threshold-free ranking quality and is the primary model-selection criterion. PR-AUC is reported alongside because, under class imbalance, it is more informative about the positive class. F1 summarises precision and recall at the default 0.50 cut-off; precision and recall are reported separately so the cost asymmetry between false positives (wasted retention spend) and false negatives (silent churn) is visible. Accuracy is reported for completeness but is unreliable here because predicting 'all non-churn' scores 71%. The full validation comparison and the final out-of-time test result are reported in **Table 2**.

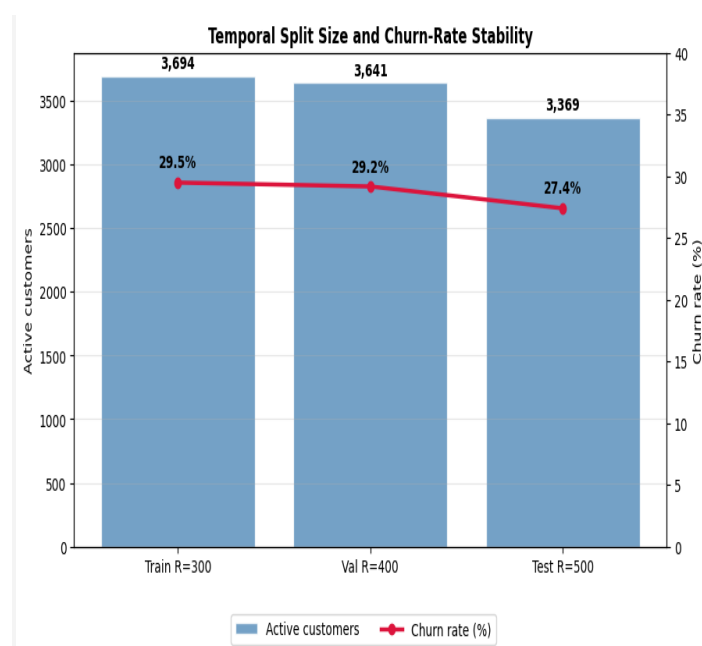


Figure 3. Temporal split sizes (3,694 / 3,641 / 3,369) and churn-rate stability across the three reference dates

Table 2 — Full model comparison (validation rows) and final test

Model	Split	Acc	Prec	Rec	F1	AUC
Baseline: all non-churn	Val	0.708	0.000	0.000	0.000	0.500
Baseline: recency > 7 days	Val	0.620	0.422	0.817	0.557	0.728
Logistic Regression (default)	Val	0.811	0.759	0.516	0.614	0.882
Decision Tree (default)	Val	0.725	0.525	0.589	0.555	0.684
Random Forest (default)	Val	0.811	0.721	0.575	0.640	0.877

HGBT (default)	Val	0.799	0.674	0.601	0.635	0.867
Logistic Regression (tuned)	Val	0.811	0.759	0.516	0.614	0.882
Random Forest (tuned) ★	Val	0.822	0.774	0.552	0.644	0.891
HGBT (tuned)	Val	0.828	0.749	0.614	0.675	0.887
Random Forest (final)	TEST	0.808	0.751	0.449	0.562	0.872

★ = model selected as final via highest validation ROC-AUC and smallest overfit gap.

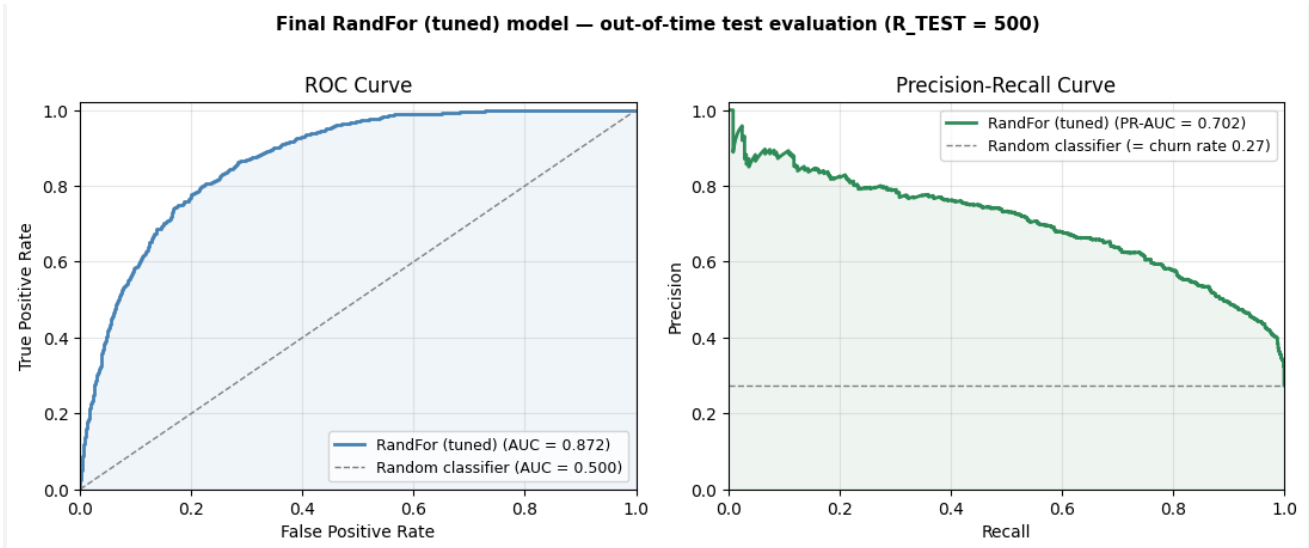


Figure 4. Final-model out-of-time test performance: ROC curve (AUC = 0.872) and Precision-Recall curve (PR-AUC = 0.702) at R_TEST = 500. The random PR baseline is 0.274 (test churn rate).

The out-of-time performance curves in Figure 4 show that the final tuned random forest retains strong ranking ability on the test reference date, with ROC-AUC = 0.872 and PR-AUC = 0.702 against a positive prevalence of 27.4%. At the default 0.50 threshold, the model achieves F1 = 0.562, precision = 0.751, recall = 0.449 and accuracy = 0.808. The confusion matrix in Figure 5 shows TN = 2,309, FP = 137, FN = 509 and TP = 414: of the 923 actual churners, 414 are captured and 509 are missed; of the 551 customers flagged, 414, or around 75%, truly churn. The drop in AUC from validation, 0.891, to test, 0.872, is 0.019, indicating modest but acceptable generalisation loss across a 100-day shift in the reference date.

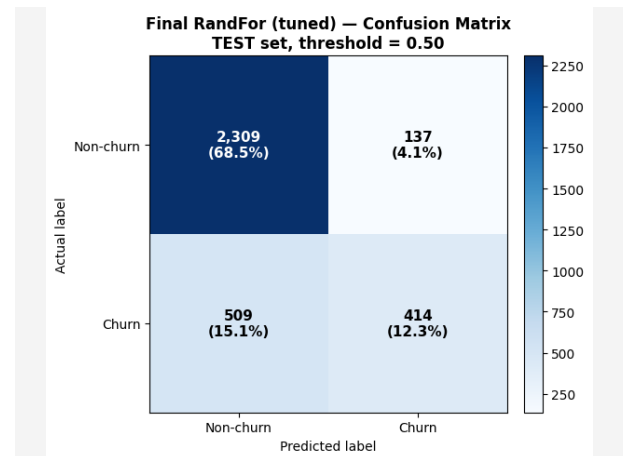


Figure 5. Final-model confusion matrix on the test set at the 0.50 default threshold.

As shown in Figure 6, the default 0.50 threshold is highly precision-favoured. A threshold sweep on the test set shows the F1 optimum at 0.28 (precision 0.621, recall 0.743, 1,105 customers flagged = 32.8% of active base); at 0.30 the trade-off is precision 0.628, recall 0.707, F1 0.665, flagging 30.9%. The correct operating point depends on intervention cost. For low-cost interventions (targeted email, app push, loyalty-points top-up) the F1-optimal threshold around 0.28–0.30 is attractive because it catches roughly 70% of true churners. For higher-cost interventions (personal call, voucher) the 0.50 default delivers 75% precision and is more cost-efficient. FoodCorp can pick the threshold operationally without retraining the model.

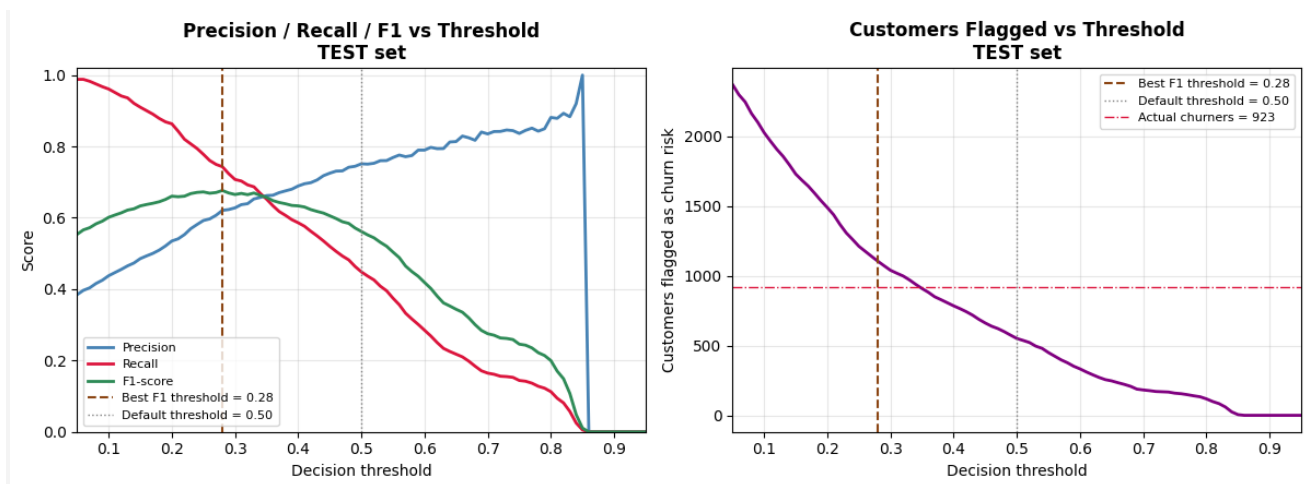


Figure 6. Threshold sweep on the test set. Precision, recall and F1 versus decision threshold; flagged-customer count versus threshold. F1 optimum at ≈ 0.28 .

Critically reflecting on usefulness: the model outperforms every baseline by a large margin (ROC-AUC 0.872 versus 0.728 for the best rule-based baseline, +0.144) and is enough to enable targeted retention. However, the single out-of-time test period leaves residual uncertainty about stability under future regime shifts (festive trading, promotional cycles, store events). Repeated temporal holdout across additional reference dates would strengthen the evidence and was identified as the principal future improvement.

3.4 Summary

The final system pairs a parameterised SQL feature pipeline with a regularised random forest trained on combined train+validation cohorts at $R = 300$ and $R = 400$, and is evaluated once on $R = 500$. Lifetime engagement features dominate the model: in the validation-set permutation importance, `total_visits_lifetime` alone delivers a 0.031 mean drop in ROC-AUC when permuted, about three times the next feature. The principal acknowledged limitations are (i) only a single test reference date, (ii) no explicit promotion-response feature, (iii) four stores only, (iv) 87% missingness in age. Section 4 details how these results inform marketing interpretation; section 5 covers the technical derivation of those insights.

4. Insight Report — Part 1: Marketing Summary

This section translates the model's findings into customer-behaviour language for the marketing team. It describes what differentiates churners from non-churners using actual differences observed in the data, and identifies behavioural signals that warrant further attention. Specific intervention design is a marketing decision and is not prescribed here.

What separates churners from non-churners. The churner and non-churner comparison in Figure 8, supported by the pen portraits in Table 3, shows that the strongest behavioural separation comes from lifetime engagement. A typical customer who subsequently churns has visited FoodCorp only twice in their entire recorded history and has spent £40 in total, while a typical retained customer has visited 15 times and spent £262 in total. The gap between consecutive shopping visits is also markedly wider for churners (mean 39.6 days, median 23 days) than for non-churners (mean 21.4 days, median 12 days). Customers showing low historical engagement therefore carry a meaningfully higher churn risk than customers with established, regular shopping patterns.

Recent behaviour adds an early-warning signal but is weaker than the lifetime picture. Customers who churn in the next 49 days have, on average, not visited for 21 days at the prediction point; non-churners have visited within 12 days. Spend in the most recent 14 days is £8.77 on average for future churners and £27.30 for future non-churners. The ratio of most-recent-window spend to prior-window spend is on average 0.09 for churners versus 0.68 for non-churners, indicating that a sharply collapsing spend trajectory is one of the clearest short-horizon flags.

Breadth of shop matters less than depth. Churners do shop across fewer distinct products (median 11) and fewer categories (median 8) than non-churners (32 and 18 respectively), but the breadth difference essentially mirrors the difference in total engagement: lower-visit customers also touch fewer SKUs. Store diversity, by contrast, is almost identical across the two groups (median 1 store each), which is unsurprising given that FoodCorp operates only four stores; this dimension is therefore not a useful behavioural signal in the current dataset.

Pen portraits. From training-set medians ($R = 300$):

Behavioural measure	Typical churner	Typical non-churner
Lifetime visits (count of distinct shopping days)	2	15

Lifetime spend (£)	40.45	261.85
Average inter-visit gap (days)	23.3	12.2
Customer tenure (days since first visit)	113	207
Days since last visit at prediction point	19	7
Visits in the most recent 14 days	0	1
Spend in the most recent 14 days (£)	0.00	14.05
Distinct products in the last 56 days	11	32
Distinct product categories in the last 56 days	8	18
Distinct stores visited in the last 56 days	1	1

Table 3 — Pen portraits (median values, training set at $R = 300$).

The contrast is most stark on lifetime visits and total lifetime spend — both differ by a factor of roughly seven between the two groups. The fact that the typical churner had only two recorded visits in their entire history with FoodCorp indicates that churn is concentrated in customers who never fully built a habit, rather than in long-standing customers who suddenly leave. This suggests that further customer research could focus on the first three to five visits as a critical onboarding window, since once customers cross that engagement threshold their behaviour begins to look like the retained group's behaviour.

The data also shows that a collapsing recent spend trajectory (visit count or value in the last 14 days well below the prior 14 days) is a useful additional short-term warning. Customers who showed visits in the prior fortnight but zero visits in the most recent fortnight are over-represented among future churners, and the model already weights this via the spend-ratio feature. This indicates that interventions tied to a recent visit gap could complement onboarding-focused activity by addressing customers who have already engaged but are now slipping.

5. Insight Report — Part 2: Technical Insight Derivation

Insights were derived in three layers: (i) univariate group comparisons between churners and non-churners on the training split only, (ii) permutation importance applied to the final tuned random forest scored on the held-out validation set, and (iii) pen portraits computed from training-set medians of the most important features. The training-only restriction is deliberate — interpreting features against the validation or test cohorts would compromise their out-of-time status.

Variable importance. The variable-importance results are shown in Figure 7. Permutation importance is used as the primary importance measure because it is model-agnostic, naturally handles correlated features and is computed on the held-out validation set rather than the training data. Each feature is shuffled ten times and the mean drop in validation ROC-AUC is recorded.

In figure 7, the five highest-importance features are `total_visits_lifetime` (0.0313), `total_spend_lifetime` (0.0113), `avg_gap_lifetime` (0.0077), `customer_tenure_days` (0.0037) and `spend_ratio_w1w2` (0.0012). The dominant signal is therefore long-run engagement; the strongest recent-behaviour feature (`spend_ratio_w1w2`, capturing the ratio of the last 14-day spend to the previous 14-day spend) is about an order of magnitude less important. `days_since_last_purchase` ranks sixth (0.0012), and individual tumbling-window features rank below it. This pattern explains why the simple recency-threshold baseline at 7 days reached $\text{ROC-AUC} = 0.728$: recency carries first-order information, but the random forest's 0.144 AUC improvement comes mainly from combining recency with the lifetime engagement context that a single recency threshold cannot express.

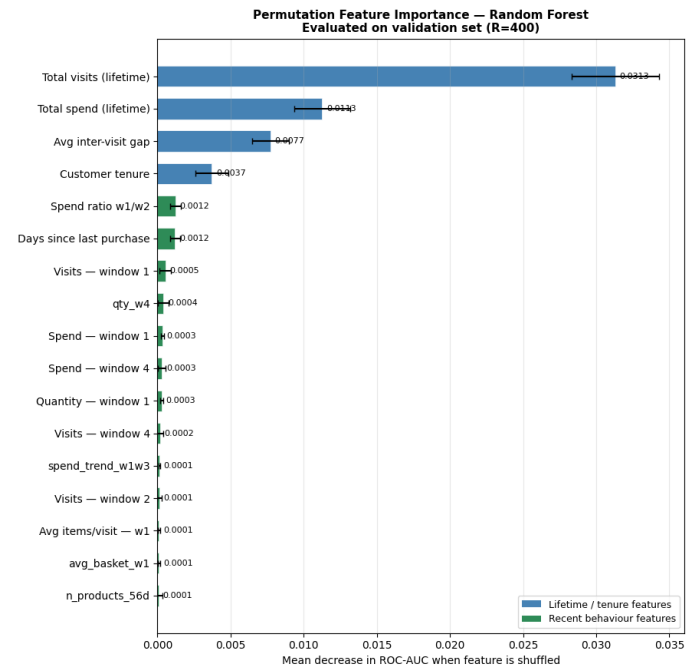


Figure 7. Permutation importance for the final random forest on the validation set, showing that lifetime engagement features dominate.

Churner vs non-churner comparisons. The group-level comparison in Figure 8 supports the same interpretation as the permutation-importance results. Mean differences between the two groups, computed on the training set only, are consistent with the importance ranking. Of the 30 retained features, the largest standardised differences are total_spend_lifetime (churners £73.00 vs non-churners £452.65, -83.9%), total_visits_lifetime (4.03 vs 25.69, -84.3%), customer_tenure_days (120.4 vs 207.2, -41.9%), n_products_56d (16.5 vs 44.3, -62.8%), and avg_gap_lifetime (39.6 vs 21.4, +85.4%). Both n_products_56d and customer_tenure_days have lower permutation importance than their raw mean differences would suggest — most likely because the random forest can capture the same signal through the cluster of correlated lifetime features rather than needing those particular columns individually.

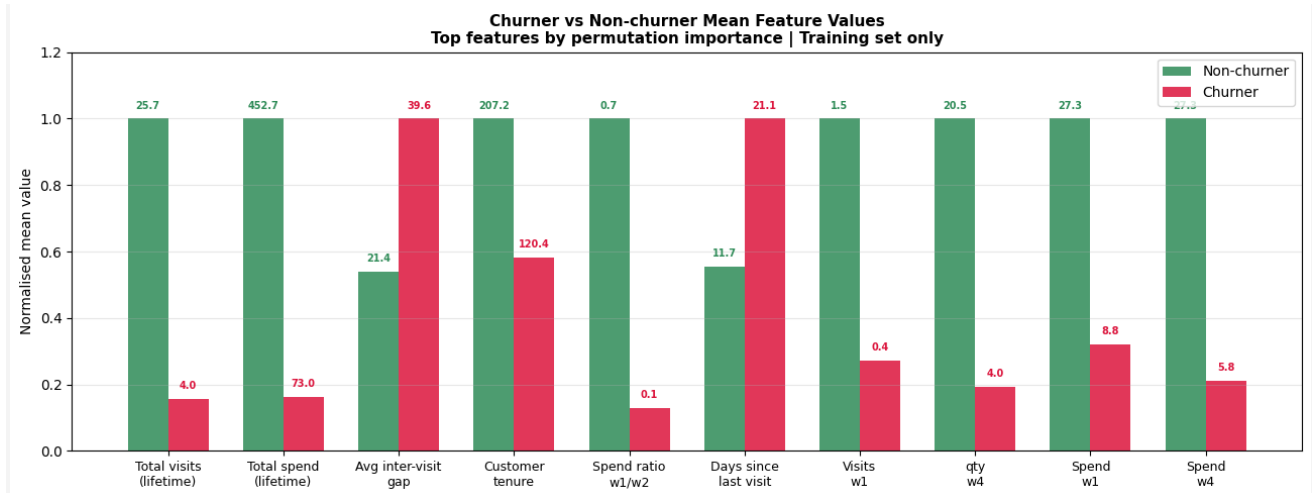


Figure 8. Normalised mean values of the top permutation-importance features for churners vs non-churners on the training set.

Pen portrait construction. Pen portraits in section 4 use medians, not means, because most behavioural features are strongly right-skewed: a small number of very high-spend customers would pull the non-churn mean upward and exaggerate the gap. Medians give a more representative picture of the typical customer in each group. They were constructed by selecting the top fourteen features by either model importance or business interpretability, computing each group's median on the training set, and grouping them into lifetime, recent-spend and breadth blocks.

Reflection on feature selection and ML pipeline. Three things worked: (i) the parameterised SQL function ensured identical feature logic across reference dates, removing a major source of leakage and drift; (ii) the pooled temporal-group scaler preserved relative magnitudes between tumbling windows; (iii) the explicit temporal holdout, with manual ParameterGrid loops scored on the validation reference date, allowed meta-parameter selection without resorting to inappropriate cross-validation. Two things were less successful: (i) age_at_R was lost to missingness, removing the only demographic, so behavioural features carry the entire signal; (ii) the single test reference date limits the strength of the stability claim. Repeated temporal holdout across additional R values would directly address (ii).

Limitations of interpretation. Permutation importance reveals which features the predictive model leans on, not which features cause churn. Low product diversity, for instance, is almost certainly a symptom of low engagement rather than an independent driver — instructing customers to buy more different products would not in itself reduce churn risk. The recency-on-recency-of-feature correlation also means importance is distributed across redundant signals; the random forest could swap which lifetime feature it relied on most without changing performance materially. Finally, the dataset gives no information about why customers churn (price, competitor openings, life events), only that they do; addressing causes requires evidence outside the transactional data.

6. Conclusion

A churn-prediction system built on parameterised SQL feature engineering, a temporal-group scaler, manual grid-search tuning on a fixed validation reference date, and a tuned random forest trained on combined train+validation cohorts achieves out-of-time ROC-AUC = 0.872 and PR-AUC = 0.702 at $R_{TEST} = 500$. This is materially better than the best rule-based baseline (ROC-AUC = 0.728) and is interpretable enough to support marketing prioritisation. Churn risk is concentrated in customers with low lifetime engagement; FoodCorp's first-three-visits onboarding period and any sharp recent-spend collapse are the two behaviour patterns most worth monitoring. Future work should extend the evaluation to a repeated temporal holdout across additional reference dates to quantify stability, add a promotion-response feature when promotion-flag data become available, and consider a separately-tuned higher-recall operating point for the model's deployment as a campaign-targeting tool.